

Total No. of Questions : 4]

SEAT No. :

PA-10288

[Total No. of Pages : 1

[6009]-322

**T.E. (Computer Engineering) (Insem.)**  
**DATA SCIENCE AND BIG DATA ANALYTICS**  
**(2019 Pattern) (Semester - II) (310251)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer questions Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is allowed.

- Q1)** a) What are dimensionality reduction and its benefits? [4]  
b) What is data wrangling? Why do you need it? [5]  
c) What is regression? Explain different types of regression with example. [6]

OR

- Q2)** a) Differentiate between Data Science, Machine Learning and AI. [4]  
b) What does feature engineering typically includes? [5]  
c) What is Data Discretization, explain Forms of data discretization. [6]
- Q3)** a) Write a short note on contingency table, explain with example. [4]  
b) With an example explain Baye's theorem. Also explain its key terms. [5]  
c) Is there a correlation between the variables in the following data set? [6]

|       |    |    |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|----|
| Hours | 9  | 15 | 25 | 14 | 10 | 18 | 19 | 16 | 20 | 18 |
| Marks | 39 | 56 | 93 | 61 | 50 | 75 | 42 | 70 | 66 | 32 |

OR

- Q4)** a) What is population & how is it differ from a sample? [4]  
b) With an example, explain one-tailed & two-tailed t-tests. [5]  
c) Describe the Chi-Square Test of Independence. [6]





**Study material provided by: Vishwajeet Londhe**

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Total No. of Questions : 4]

SEAT No. :

**PB-103**

[Total No. of Pages : 2

**[6269]-317**

**T.E. (Computer Engineering) (Insem)**

**DATA SCIENCE AND BIG DATA ANALYTICS**

**(2019 Pattern) (Semester - II) (310251)**

**Time : 1 Hour]**

**[Max. Marks : 30**

**Instructions to the candidates:**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

- Q1) a)** Explain data wrangling methods with suitable example. **[5]**
- b) Suppose that the data for analysis includes the attribute age, given the following data (in increasing order) for the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. **[5]**
- i) Use smoothing by bin means, using a bin depth of 3.
  - ii) What other methods are there for data smoothing?
- c) What is data science? Compare data science and information science. **[5]**

**OR**

- Q2) a)** Explain 5 V's of Big Data. **[5]**
- b) Explain different phases of data analytics life cycle with neat diagram. **[5]**
- c) Compare Business Intelligence and data science. **[5]**
- Q3) a)** Explain skewness and kurtosis. What is the purpose of finding skewness of data? **[5]**
- b) What is degree of freedom? Explain with example. **[5]**
- c) How hypothesis testing works? Explain steps. **[5]**

**OR**

**P.T.O.**

- Q4)** a) List out measures of dispersion with their significance and mathematical formulae. [5]
- b) Describe Chi-square Goodness of Fit test. [5]
- c) Assume that a patient X took a lab test for a certain disease and tested positive. The lab test returns a positive result in 95% of the cases in which the disease is actually present and it falsely returns a positive result in 6% of the cases in which the disease is not present. Further more only 1% of the entire population has this disease. What is the probability that X actually has the disease given that he is tested positive. [5]



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SEAT No. :

**PD96**

[Total No. of Pages : 2

**[6410]-417**

**T.E. (Computer Engineering) (Insem)**  
**DATA SCIENCE AND BIG DATA ANALYTICS**  
**(2019 Pattern) (Semester - II) (310251)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

- Q1)** a) What is data science? Compare data science and information science. [5]
- b) What are the steps of data preprocessing? What steps may the change the dimensions of the data? [5]
- c) What is data discretization? What are the different techniques used for data discretization? [5]

OR

- Q2)** a) Describe different ways of handling missing data with suitable example. [5]
- b) In urban planning applications, devise the problems that can be solved using data science. [5]
- c) List and explain data reduction techniques with example. [5]

- Q3)** a) For different levels of measurements like nominal, ordinal, interval and ratio, write the measures (among mean, median and mode) that can be used? [5]
- b) The table below shows the weights of members (in kg) in a sports club. Calculate mean, mode and median of the distribution. [5]

|           |       |       |       |       |       |       |
|-----------|-------|-------|-------|-------|-------|-------|
| Weights   | 40-49 | 50-59 | 60-69 | 70-79 | 80-89 | 90-99 |
| Frequency | 6     | 8     | 12    | 14    | 7     | 3     |

- c) Explain the steps of hypothesis testing with suitable example. [5]

OR

**P.T.O.**

- Q4)** a) What is normal distribution and skewed distribution? What is the relation between mean, mode and median values in both the cases? [5]
- b) List out measures of dispersion with their significance and mathematical formulae. [5]
- c) Imagine a company wants to test the claim that their batteries last more than 40 hours. Using a simple random sample of 15 batteries yielded a mean of 44.9 hours, with a standard deviation of 8.9 hours. Calculate t-value (one-sample t-test) Test this claim using a significance level of 0.05 and P value = 0.026. [5]

